Report: 3D Structure Prediction of the ACMV Capsid Protein

CodeAlpha Internship, Bioinformatics, Task 3

Author: Pascal Houngue

Date: October 21, 2026

1. Introduction

African cassava mosaic virus (ACMV) infects cassava, causing significant yield losses in Africa. Its capsid protein forms the viral envelope, protecting the viral DNA. It also plays a role in the transmission of the virus by whiteflies.

In this task, I predict the 3D structure of this protein, visualize it using ChimeraX, and describe its structural characteristics.

2. Selected Protein

I have chosen the capsid protein of ACMV. Its UniProt code is P03561 (name: CAPSD_CLVK). It consists of 258 amino acids. Its entry is reviewed (Swiss-Prot).

I have already used this protein in Tasks 1 and 2; therefore, I am maintaining the same workflow: sequence, comparison, and then structure.

3. Method

1. I consulted the UniProt entry, which indicates an experimental structure: 6EK5. This structure was obtained via cryo-electron microscopy and has a resolution of 4.2 Å. It covers positions 48 to 252.

2. I searched for the protein in the AlphaFold DB but found no entry for P03561.

3. Consequently, I used SWISS-MODEL. The software selected the template 6ek5.1.A, which shares 99.02% sequence identity with my protein.

4. I opened the model in UCSF ChimeraX (version 1.12), generated images, and counted the structural elements.

5. I superimposed the model onto the 6EK5 structure (chain A) using the `matchmaker` command.

4. Results

4.1 Model quality

The model is a monomer. It covers 190 out of 258 residues. Its scores are as follows:

Metric	Value
Identity with template (6EK5)	99,02 %
GMQE	0,54
Global QMEANDisCo	0,58 ± 0,06
Modeled residues	190 sur 258

The scores are average. There are two reasons for this: the model covers only a portion of the protein, and the template has a medium resolution (4.2 Å).

4.2 Structural characteristics

The model consists primarily of β-sheets. It contains very few helices.

Element	Residues	Share of the model
β-sheets	70	approximately 37%
Helices	6	approximately 3%
Loops and irregular regions	114	approximately 60%

The β-sheets form a compact core. The loops are located at the periphery. The loops connect the strands to one another. They are the most flexible regions.

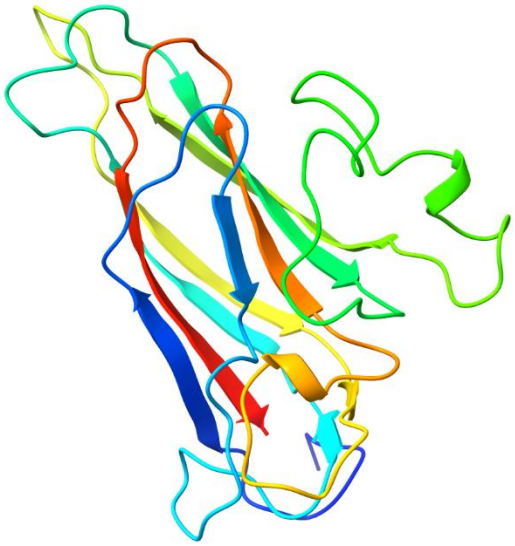


Figure 1. Overview of the model. The color ranges from blue (beginning) to red (end).

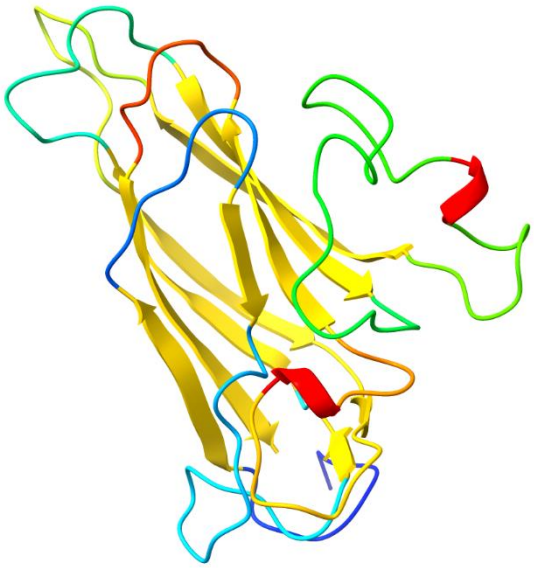


Figure 2. Secondary structure. β-sheets are in yellow. Helices are in red.

4.3 Comparison with the experimental structure

I superimposed my model onto chain A of 6EK5. The RMSD is 0.162 Å, calculated over 183 pairs of residues. The two structures are nearly identical. Minor differences are found in the loops.

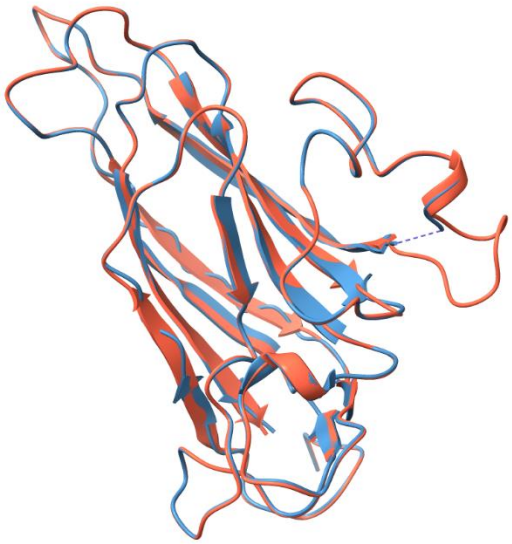


Figure 3. Superposition of the model (red-orange) and 6EK5, chain A (blue).

5. Discussion

The protein features a fold rich in β-sheets. This fold is consistent with that of a viral capsid protein. Indeed, the experimental structure 6EK5 shows the protein within its complete capsid.

The very low RMSD was expected, as SWISS-MODEL constructed the model based on 6EK5. This result therefore demonstrates that the model closely follows its template; it does not constitute an independent validation.

6. Limitations

- The model represents a single monomer, whereas the actual capsid contains multiple copies of the protein.
- The model does not cover the remaining 68 residues, primarily at the beginning of the sequence; these are also absent from the experimental structure.

- The resolution of 6EK5 is moderate (4.2 Å), meaning fine details are less precise.
- The GMQE and QMEANDisCo scores are estimates and do not replace experimental measurements.
- The secondary structure percentages are derived from automated assignment by ChimeraX and are therefore approximate.

7. Conclusion

I predicted the 3D structure of the ACMV capsid protein using SWISS-MODEL and visualized the resulting model in ChimeraX. The structure is dominated by β-sheets and loops, and it superimposes almost perfectly onto the experimental structure 6EK5. This work illustrates the process of moving from a sequence to a 3D structure and subsequently interpreting it.

Tools used

- UniProt (entry P03561)
- AlphaFold DB (no entry found)
- SWISS-MODEL
- UCSF ChimeraX, version 1.12
- Experimental PDB structure: 6EK5